FICTIONAL TEST CV - FOR SOFTWARE TESTING ONLY

# MINSEO KIM

Undergraduate Researcher | Computer Vision, Robotics, and AI

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## PROFILE

Computer science undergraduate interested in intelligent robotic systems that perceive, learn from demonstrations, and adapt to real-world environments. Experienced with computer vision, 3D geometry, deep learning, backend APIs, and reproducible experimentation. Seeking graduate research opportunities in robotics, human-robot interaction, embodied AI, or vision-based manipulation.

## RESEARCH INTERESTS

Robotic manipulation; learning from demonstration; computer vision; 3D reconstruction; human-robot interaction; embodied AI; vision-language models; trajectory generation; depth sensing; explainable recommendation systems.

## EDUCATION

**B.S. in Computer Science and Engineering** | *Hanbit University* — **Mar 2022 - Feb 2027 (expected)**

Seoul, Republic of Korea

- GPA: 4.08/4.50; Major GPA: 4.21/4.50.
- Relevant coursework: Artificial Intelligence, Machine Learning, Computer Vision, Computer Graphics, Database Systems, Operating Systems, Algorithms, Computer Architecture, and Probability & Statistics.
- Capstone focus: AI-assisted discovery of graduate research laboratories and explainable professor-lab matching.

## RESEARCH EXPERIENCE

**Undergraduate Research Assistant** | *Intelligent Systems Laboratory, Hanbit University* — **Sep 2025 - Present**

Seoul, Republic of Korea

- Developed a learning-from-demonstration pipeline for a 6-DoF robot arm performing kitchen manipulation tasks, including data collection, trajectory resampling, conditional generation, inverse kinematics, and safe execution.
- Built computer-vision modules using RGB-D cameras, ArUco markers, OpenCV, and coordinate transforms to estimate workspace geometry and express trajectories in an object-centered frame.
- Trained 1D CNN and conditional autoencoder models in PyTorch for action classification and trajectory generation; achieved 100% test accuracy on a controlled 10-action dataset.
- Designed evaluation scripts for positional error, rotational error, joint discontinuity, and reproducible generation across repeated runs.

**Student Research Intern** | *Robotics and Perception Group, Mirae Institute of Technology* — **Jun 2026 - Aug 2026**

Pohang, Republic of Korea

- Investigated pot-grounded trajectory generation for robotic cooking using an eye-to-hand ZED 2i depth camera and a UFactory Lite6 robot arm.
- Calibrated camera-to-robot transforms and validated the system with translation RMS of 2.7 mm and rotation RMS below 1 degree in a controlled setup.
- Implemented a prototype for recovering top-view geometry from oblique depth observations and explored automatic estimation of rim, bottom plane, radius, and height.

## SELECTED PROJECTS

**LabMatch AI - Explainable Graduate Lab Recommendation Platform** | *Team Project* **Jul 2026**

FastAPI, PostgreSQL, React, OpenAI API

- Created a web service that analyzes a student CV and recommends professors and laboratories using keyword matching, embedding similarity, recent-paper relevance, user preferences, and data freshness.
- Defined an explainable scoring output with total score, matched and missing keywords, score breakdown, evidence, concise rationale, and recommended next action.
- Implemented REST API contracts for CV upload, profile extraction, recommendation retrieval, saved laboratories, and application scheduling.

**Pot-Grounded Robotic Cooking Dataset and Policy Pipeline** | *Individual Research Project* **Jul 2026**

Python, PyTorch, OpenCV, NumPy

- Collected and processed 90 direct-teaching trajectories across 10 cooking action variants, with each trajectory normalized to 300 time steps.
- Built a complete pipeline from joint-space data to pose conversion, object-centered normalization, conditional trajectory generation, inverse kinematics, upsampling, and robot execution.
- Produced reproducible experiment metadata, plots, model checkpoints, and command-line tools for dry-run validation and deployment.

## TECHNICAL SKILLS

**Programming:** Python, C, C++, Java, SQL, JavaScript/TypeScript, Bash

**Machine Learning:** PyTorch, scikit-learn, CNNs, autoencoders, embeddings, model evaluation

**Computer Vision & Robotics:** OpenCV, RGB-D cameras, ArUco, camera calibration, coordinate transforms, forward/inverse kinematics, trajectory processing

**Backend & Data:** FastAPI, REST APIs, PostgreSQL, MySQL, SQLAlchemy, Pydantic, JWT basics

**Development Tools:** Git, GitHub, Docker, Linux/WSL, VS Code, Jupyter, CMake, pytest

## PUBLICATIONS AND PRESENTATIONS

**Object-Centered Trajectory Generation for Robotic Stir-Fry Manipulation** | *Undergraduate Research Symposium* **Nov 2026 (planned)**

Hanbit University

- Poster abstract in preparation. This entry is fictional and included only to test publication and date extraction.

**A Multi-Signal Framework for Explainable Research Laboratory Recommendation** | *Project Technical Report* **Jul 2026**

LabMatch AI Team

- Documented a weighted recommendation method combining lexical, semantic, publication, preference, and freshness signals.

## AWARDS AND ACTIVITIES

**Open Innovation Build Week** | *Education Track Participant* **Jul 2026**

Remote / Team of four

- Developed a functional prototype connecting a React frontend, FastAPI backend, database, CV analysis, laboratory crawler, recommendation engine, and contact-email assistant.

**Academic Excellence Scholarship** | *Hanbit University* **2024 - 2026**

Seoul, Republic of Korea

- Awarded for strong academic performance in three consecutive semesters.

**Teaching Assistant** | *Data Structures Laboratory* **Mar 2025 - Jun 2025**

Hanbit University

- Supported weekly programming labs, reviewed student code, and explained debugging, complexity, and version-control fundamentals.

## ADDITIONAL INFORMATION

**Languages:** Korean (native), English (intermediate working proficiency)

**Research preferences:** Robotics or AI laboratory; South Korea preferred; open to interdisciplinary programs; interested in advisors publishing actively in robot learning, 3D vision, and human-centered AI.

**Availability:** Graduate study from Spring 2027; available for remote research collaboration before enrollment.